



2D construction planning for swarms of simple earthmover robots

Jiahe Chen¹ · Kirstin Petersen¹

Received: 16 May 2025 / Revised: 25 September 2025 / Accepted: 22 October 2025 / Published online: 15 November 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

New settlements in remote environments require terrain modification, a task well suited for autonomous multi-robot systems. Simple, robust earthmover robots offer an inexpensive and scalable alternative to sophisticated construction robots. We present a mathematical model for such robots modifying continuous granular structures in 2D and develop both centralized and decentralized planning algorithms to achieve user-defined construction goals. These algorithms decompose long-horizon tasks into subtasks solvable using optimal transport theory and Wasserstein geodesics. Simulations across 100 randomly generated tasks show that a centralized controller with global information achieves on average 85% and 92% construction progress on untraversable and traversable terrains respectively, even with action noise. Multiple robots reduce overall travel distance by 70%, important because motion over the structure also disturbs it. The distributed algorithm—without global information—matches centralized performance on traversable terrain, reaching 93% progress. Increasing robot numbers accelerates convergence, lowers moved material, and raises convergence rates, though congestion can increase total travel distance. These results indicate that simple earthmover robots hold promise for construction tasks ranging from extraterrestrial habitat preparation to coastal protective berms.

Keywords Robotics and automation in construction · Distributed robot systems · Multi-robot systems · Optimal transport theory

1 Introduction

Robotics is often envisioned as a means to prepare remote, hostile landscapes for colonization, whether in the Arctic, the deep sea, or outer space. Government agencies and private investors planning an imminent return to the moon and future missions to Mars are driving innovations that call for robots capable of using in-situ materials to build landing pads, roads, foundations, and berms (Desai et al., 2022; Moses & Mueller, 2021). Current approaches typically rely on advanced actuation, high-speed communication, and sophisticated sensors (Lee et al., 2023; Hong et al., 2022; Thangavelautham & Xu, 2022; Hong et al., 2021). Yet the extreme conditions of extraterrestrial terrain modification

create high risks of clogged mechanisms, dust-covered or saturated sensors, and robot entrapment; long-term resilience and autonomy are therefore essential. A promising alternative is to deploy swarms of simpler, more robust robots that can continue working even when individual units fail (Cheraghi et al., 2022).

In earlier work (Huang et al., 2022), we introduced and experimentally tested such a minimalist robot consisting of a double-tailed single-actuator wave mechanism (Fig. 1a). Fixed mechanical helices surrounded by rectangular chains generate a sinusoidal motion capable of traversing diverse terrains (Shachaf et al., 2019; Zarrouk et al., 2016). By driving the two tails at different speeds, the robot can push, transport, level, compact, and dig into granular materials, as well as climb over other robots. These demonstrations, using tethered robots to manipulate large granules (packing peanuts) in 2D, highlighted the robustness of these construction behaviors, many of which required only open-loop control. In this paper, we build on this technology by introducing an abstract mathematical model for the robot mechanism and multi-robot planning and coordination algorithms

✉ Jiahe Chen
jc3472@cornell.edu
Kirstin Petersen
kirstin@cornell.edu

¹ School of Electrical and Computer Engineering, Cornell University, 136 Hoy Rd, Ithaca, NY 14853, USA

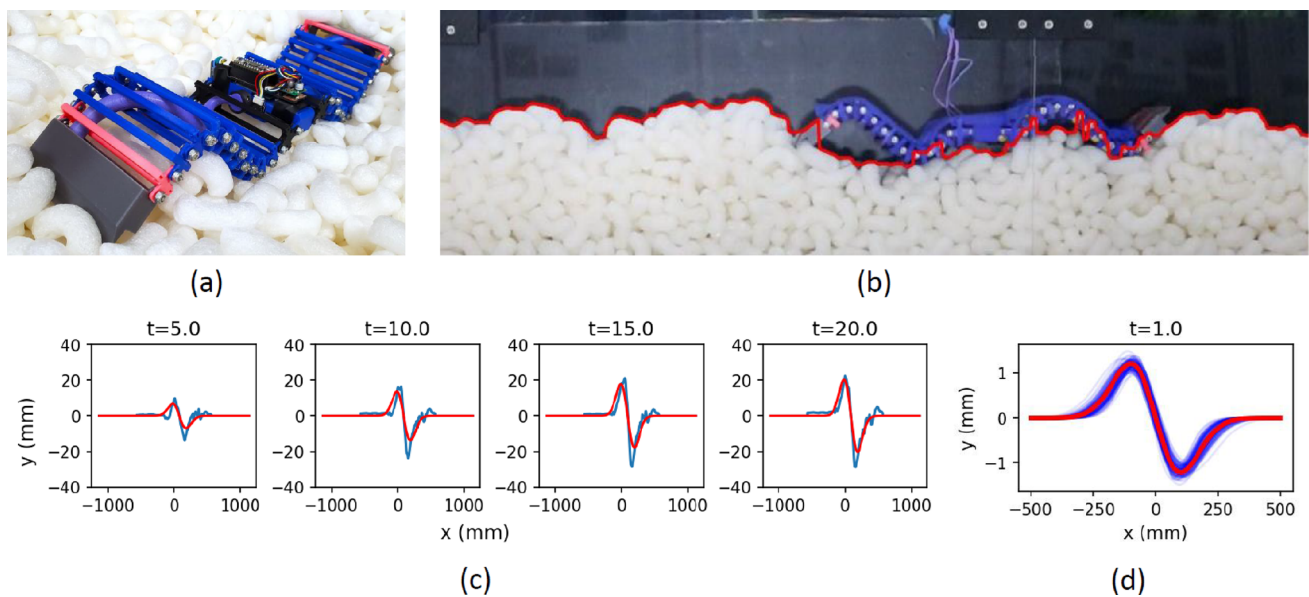


Fig. 1 The construction function (Eq. 2) is obtained through curve fitting with real data. **a** A double-tailed SAW robot on granules. **b** Automated surface tracking of a SAW robot pushing material forward. **c**

Fitting the change of surface over time (blue line) with the construction function (red line). **d** Construction function without noise (red line) and with noise (blue line)

based on Wasserstein geodesics, also aptly known as the Earth Mover's Distance.

Collective robotic construction has emerged as a distinct field with unique challenges and opportunities (Petersen et al., 2019). Prior work on planners spans a wide range of goal structures, materials, and platform constraints. The most closely related approach introduced navigability constraints so simple robots could modify terrain guided only by their ability to overcome discontinuities and slopes (Napp & Nagpal, 2014), later extended to diverse materials such as foam bricks, sandbags, and rocks (Saboia et al., 2019; Thangavelu et al., 2020). Our work similarly targets continuous amorphous structures but focuses on non-additive construction: agents cannot carry material from elsewhere but must push existing material to form a target shape. Compared with Napp and Nagpal (2014), our problem accepts arbitrary initial structures and requires reaching a user-defined goal, making the task both broader in input and stricter in output and thus more challenging to model and plan.

Our methods represent a first step toward landscape preparation with simple, robust autonomous multi-robot systems. Central to this contribution is an abstract model of non-additive terrain modification (Sect. 2) coupled with a planning and control framework based on Wasserstein geodesics (Sect. 3). This single mathematical foundation supports both centralized and distributed multi-robot strategies (Secs. 4–5), allowing us to compare their performance and highlight the trade-offs between global coordination and local autonomy (Sect. 6). The work in this article is an extension of the work presented at the 17th International

Symposium for Distributed Autonomous Robotic Systems (DARS'17): the distributed algorithm and comparison to the centralized algorithm is unique to this paper.

Our results advance the vision of deploying minimalist robot swarms to prepare extraterrestrial or otherwise inaccessible environments, e.g. by shaping landing sites, berms, or access paths (Desai et al., 2022; Moses & Mueller, 2021), and suggest that the same optimal-transport principles could be applied to other tasks involving manipulation of continuous structures, such as deformable object shaping (Ondras et al., 2022).

2 Abstract model

The construction problem is formulated as follows. Given initial and user-defined goal structures, how can we design a robot motion and action strategy such that the shape of the final structure is as close to the goal structure as possible? The real robot has two construction modes. In one mode, one tail is stationary and keeps the robot in place while the other moves material forward. In the other mode, locomotion and material manipulation happen simultaneously. Here, we focus on the former one to separate locomotion and manipulation. The remainder of this section presents mathematical models of the continuous structure and modifications; the height function and function operator concepts are inspired by Napp and Nagpal (2014).

2.1 Continuous structure modeling

We model the 2D construction similar to the original robot environment (Huang et al., 2022), also shown in Fig. 1b. The construction area Q can be any connected, compact, and finite subset of $\mathbb{R}^1$. Here we set it as an area with length L , $Q = [-L/2, L/2]$. The height function is a continuous, bounded, non-negative function, $h : Q \rightarrow \mathbb{R}^+$. The graph of h , $(x, h(x))$ with $x \in Q$ describes a structure. Due to the physical properties of granular material, the maximum steepness of the structure cannot exceed a constant $K = 0.42$, informed by real robot experiments (Huang et al., 2022). Therefore, the height of a valid structure must satisfy:

$$\max |h'(x)| \leq K \quad (1)$$

2.2 Construction function and noise modeling

The robot action on the structure can be modeled by a class of real-valued bounded construction functions $f : Q \rightarrow \mathbb{R}$. The integral of a valid construction function over Q must be 0 so that the volume of the structure is preserved: $\int_Q f(x) dx = 0$. We use the first Gaussian derivative to model a robot pushing material. Let $Q_f \subset Q$ denote the area where the action can be taken. Given an action location $\phi \in Q_f$, and a direction of pushing the material $\theta \in \{\text{left}, \text{right}\}$, the construction function of pushing material for 1 s is:

$$f_{\phi, \theta}(x) = \frac{z(\theta)a}{\sqrt{2\pi}w^2} (x - \phi) e^{-\frac{(x-\phi)^2}{2w^2}}, \quad (2)$$

where $a \neq 0$ and $w > 0$ control the size of the moved mass, and $z(\text{right}) = 1, z(\text{left}) = -1$. a and w are set to fit the real robot behavior. The fitting process is shown in Fig. 1b–d. The area where the action is permitted is set to $Q_f = [-(L/2 - 5w), (L/2 - 5w)]$, so that $f_{\phi, \theta}$ is always a valid construction function (assuming $L \gg 10w$).

In reality, the robot's action is always noisy. We add Gaussian noise to the shape parameters and action position to mimic this effect. The construction function with noise is:

$$\hat{f}_{\phi, \theta}(x) = \frac{z(\theta)\hat{a}}{\sqrt{2\pi}\hat{w}^2} (x - \hat{\phi}) e^{-\frac{(x-\hat{\phi})^2}{2\hat{w}^2}}, \quad (3)$$

where the noisy parameters are drawn from normal distributions:

$$\hat{a} \sim \mathcal{N}(a, a/10), \quad \hat{w} \sim \mathcal{N}(w, w/10), \quad \hat{\phi} \sim \mathcal{N}(\phi, 10).$$

It is important to note that the deterministic parameters (a, w) are derived from the curve-fitting process, while the noise variances instead were calibrated empirically from repeated robot trials. The observed variability in displacement was approximated by 10% of the mean values of a and w , and a ± 10 unit perturbation in ϕ . This separation ensures that the base model captures the average displacement behavior, while the stochastic extension realistically reflects execution uncertainty. Although the noise variance may appear small per action, repeated application accumulates into significant deviations over long horizons, mirroring real-world robot performance.

2.3 Modification modeling

Let $\mathcal{H}$ denote the space of all valid height functions on Q . Let $\mathcal{F}$ denote the space of all valid construction functions. The modification process is modeled by the modification operator $M : \mathcal{H} \times \mathcal{F} \rightarrow \mathcal{H}$. Let parentheses $(\cdot)$ denote the function application to points and brackets $[\cdot]$ denote the operator application to functions. Given a current structure h and a construction function f , the new structure after modification is:

$$M[f, h](x) = h(x) + \alpha \hat{f}(x), \quad (4)$$

where $\alpha \in [0, 1]$ is an attenuation factor that enforces the steepness constraint of the structure (Eq. 1) and $\hat{f}$ is the construction function with noise (Eq. 3). In reality, when the robot keeps digging at the same place and the steepness of the hole exceeds K , the surrounding material will roll back. In addition, the robot has limited weight and cannot displace material heavier than its own weight. Therefore, the robot action has a diminishing effect when the steepness of the modified structure approaches K , which is captured by α . Let $x^* = \operatorname{argmax} |h'(x) + \hat{f}'(x)|$. α is given by:

$$\alpha = \begin{cases} 1 & \text{if } |h'(x^*) + \hat{f}'(x^*)| \leq K \\ \frac{K - h'(x^*)}{\hat{f}'(x^*)} & \text{if } h'(x^*) + \hat{f}'(x^*) > K \\ \frac{-K - h'(x^*)}{\hat{f}'(x^*)} & \text{if } h'(x^*) + \hat{f}'(x^*) < -K \end{cases} \quad (5)$$

Given an initial structure h_0 , and a sequence of actions characterized by the position and direction parameters $(\phi_1, \theta_1), (\phi_2, \theta_2), (\phi_3, \theta_3) \dots$, the final structure after n actions is:

$$M[f_{\phi_n, \theta_n}, M[f_{\phi_{n-1}, \theta_{n-1}}, \dots, M[f_{\phi_1, \theta_1}, h_0]]] = h_0(x) + \sum_{i=1}^n \alpha_i \hat{f}_{\phi_i, \theta_i}(x) \quad (6)$$

2.4 Robot constraint modeling

For a real robot, there is a maximum steepness $K_R = 0.3$ it can climb before becoming stuck (Huang et al., 2022) (it can still descend a slope with steepness $< -K_R$). After imposing this constraint, the robot will not travel to any location outside the reachable area. Given the x-coordinate of robot location x_R , the reachable area $Q_R \subset Q$ is the largest closed interval $[a, b]$ that includes x_R and satisfies following conditions:

$$\begin{aligned} h'(x) &\leq K_R \quad \text{for } b \geq x > x_R, \\ h'(x) &\geq -K_R \quad \text{for } a \leq x < x_R, \\ \partial_+ h(x_R) &\leq K_R \quad \text{or} \quad \partial_- h(x_R) \geq -K_R, \end{aligned} \quad (7)$$

where $\partial_+ h(x_R)$ and $\partial_- h(x_R)$ are the right and left derivative at the robot location. If the robot receives a command to go to a location outside Q_R , it will simply ignore it. Notice that after a robot travels to a location in Q_R , it may not be able to go back to its previous location. This constraint makes the construction process much more unpredictable as new actions can unintentionally change the accessible area for the robot.

3 Planning algorithm

Based on Eq. 6, we can reformulate the construction problem as follows. Given an initial structure h_0 and a goal structure h_1 which must have the same integral over Q , how can we find a sequence of actions $\{f_{\phi_i, \theta_i}\}$ that can modify h_0 into a structure as close as possible to h_1 ? In this section, we present a planner that automatically abstracts the construction problem into multiple subtasks, which are much easier to solve based on Wasserstein geodesics. In each subtask, the algorithm generates an intermediate goal structure h_t based on the current structure, and then uses a greedy approach to find an action sequence that can modify the current structure to h_t . The remainder of this section reviews optimal transport theory and Wasserstein distance, and then presents the basic framework of the planning algorithm.

3.1 Optimal transport and wasserstein metric

Optimal transport theory was originally formulated by Monge to study the most efficient way to transform a distribution of mass to another (Monge, 1781). In recent years, it has been widely used in statistics, machine learning, and image processing (Arjovsky et al., 2017; Lellmann et al., 2014; Ramdas et al., 2017). Here we briefly introduce optimal transport theory and the Wasserstein metric without going into details of measure theory and Riemannian

geometry. While our current focus is on 2D terrain modification, which corresponds to the Wasserstein metric in 1D, we will begin with the general formulation in n D to lay the groundwork for deriving the 1D case and presenting future extensions to 3D construction.

Consider two probability density functions I_0 and I_1 defined over $\Omega \subset \mathbb{R}^n$. A transport map $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ tells where to move the mass from one location to another so that I_0 can morph into I_1 . For a valid transport map, the amount of mass moved out of an area must be the amount of mass moved into another area. Therefore, the measure-preserving (MP) requirement must be met:

$$\int_{T^{-1}(A)} I_0(x) dx = \int_A I_1(x) dx. \quad (8)$$

If T is differentiable and bijective, by applying a change of variables, we have:

$$\det(DT^{-1}(x)) I_0(T^{-1}(x)) = I_1(x), \quad (9)$$

where $\det(DT^{-1}(x))$ is the Jacobian of T^{-1} . If the cost of moving a unit mass is measured by a convex distance function $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, then the optimal transport problem based on the Monge formulation is:

$$M(I_0, I_1) = \inf_{T \in MP} \int_{\mathbb{R}^n} d(x, T(x)) I_0(x) dx. \quad (10)$$

The minimizer T^* is called the optimal transport map.

The 2-Wasserstein distance in n D is defined as:

$$W_2(I_0, I_1) = \left(\int_{\mathbb{R}^n} |x - T^*(x)|^2 I_0(x) dx \right)^{1/2}. \quad (11)$$

Let $P(\Omega)$ be the set of probability densities supported on Ω , then the metric space $(P(\Omega), W_2)$ is referred to as the 2-Wasserstein space. The 2-Wasserstein geodesics $I_t, t \in [0, 1]$ is the shortest path connecting I_0 and I_1 in 2-Wasserstein space:

$$I_t(x) = \det(DT_t^{-1}(x)) I_0(T_t^{-1}(x)), \quad T_t(x) = (1-t)x + tT^*(x). \quad (12)$$

T^* in higher dimensions can be estimated using numerical methods such as flow minimization, gradient descent, linear programming, and entropy regularization (Angenent et al., 2003; Chartrand et al., 2009; Cuturi, 2013; Oberman & Ruan, 2015).

In the case of 1D, the optimal transport map, 2-Wasserstein distance, and 2-Wasserstein geodesics all have closed-form solutions.

$$T^*(x) = F_1^{-1} \circ F_0(x), \quad (13)$$

$$W_2(I_0, I_1) = \left(\int_0^1 |F_0^{-1}(t) - F_1^{-1}(t)|^2 dt \right)^{\frac{1}{2}}, \quad (14)$$

$$I_t(x) = \frac{I_0(T_t^{-1}(x))}{T_t'(T_t^{-1}(x))}, \quad T_t(x) = (1-t)x + tT^*(x), \quad (15)$$

where F_0 and F_1 are cumulative distribution functions of I_0 and I_1 .

To move along the Wasserstein Geodesics, we refer to Eq. 11 which also suggests that the 2-Wasserstein geodesics has the following properties:

$$W_2(I_0, I_t) = tW_2(I_0, I_1). \quad (16)$$

This means if we start from I_0 and move towards I_1 by a distance dW along the geodesics, we can set $t = dW/W_2(I_0, I_1)$ and use this value to find the corresponding intermediate structure I_t as our target.

3.2 Measure of distance between structures

We apply the Wasserstein metric in 1D to both measure the distance between structures and generate intermediate structures. The height function h is converted to a probability distribution function I by dividing by the integral over Q . Consider two arbitrary height functions h_a and h_b with the same integral v over Q . Let $\mathcal{Q}$ denote the space of real-valued, bounded functions on Q and let $\mathcal{Q}^+$ denote the non-negative subset of $\mathcal{Q}$. Thus we have $\mathcal{H} \subset \mathcal{Q}^+ \subset \mathcal{Q}$. Eq. 14 can be used to measure the distance between them. Let $D_{W_2} : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$ be the distance function based on 2-Wasserstein distance. Let $D_{L_2} : \mathcal{Q} \times \mathcal{Q} \rightarrow \mathbb{R}$ be the distance function based on Euclidean distance.

$$D_{W_2}(h_a, h_b) = W_2(h_a/v, h_b/v), \quad (17)$$

$$D_{L_2}(h_a, h_b) = \left(\int_Q |h_a(x) - h_b(x)|^2 dx \right)^{\frac{1}{2}}. \quad (18)$$

For evaluation purposes, we also need a measure of progress. Given the initial structure h_0 , goal structure h_1 , and current structure h , the construction progress is $1 - D_{W_2}(h, h_1)/D_{W_2}(h_0, h_1)$.

Finally, to generate an intermediate structure at a point t between two given structures h_a and h_b , we construct an operator $G_t : \mathcal{Q}^+ \times \mathcal{Q}^+ \rightarrow \mathcal{Q}^+$ based on Eq. 15.

$$G_t[h_a, h_b](x) = \frac{h_a(T_t^{-1}(x))}{T_t'(T_t^{-1}(x))}, \quad T_t(x) = (1-t)x + tF_b^{-1} \circ F_a(x), \quad (19)$$

where F_a, F_b are cumulative distribution functions of h_a and h_b . Notice that T_t^{-1} always exists since all height functions are non-negative and continuous.

3.3 Planning algorithm based on wasserstein geodesics

To introduce and evaluate the planner for single robot construction and centralized multi-robot construction, we make several assumptions. We assume that robots have global sensing and communicate with a central controller without delay. We also assume that the robot movement does not change the structure's shape, but we discuss how to alleviate this issue later.

The basic single-robot planning algorithm is shown in Alg. 1. The algorithm decomposes the construction problem into subtasks, in which it computes the shortest path (geodesics) from the current to the goal structure and then attempts to move the structure by dW over that path (lines 3–5). If the distance to the goal is less than dW , the algorithm attempts to move towards the goal directly. The algorithm then uses a greedy approach to determine the action sequence that leads to the intermediate goal. Notice that since the prediction process does not capture the noise, the attenuation factor, and robot constraints, the execution result (line 14) might be very different from the predicted outcome.

Notice that during the greedy search (line 9), we use Euclidean distance D_{L_2} instead of Wasserstein distance D_{W_2} . The reason is as follows. Compared with D_{L_2} , D_{W_2} captures the geometry of the data and can more accurately measure the progress towards the goal (Kolouri et al., 2017) (also illustrated in Fig. 2). However, D_{W_2} is more expensive to compute, especially in higher dimensions, as shown in Table 1. In Alg. 1, most of the computational effort is concentrated on the greedy search step (line 9). For this reason, we employ D_{L_2} rather than D_{W_2} at this step. Since the greedy search requires evaluating the distance at every candidate location, this computational gap is amplified by the number of locations. Moreover, when the objective is to reach an intermediate structure that differs only slightly from the current one, D_{L_2} and D_{W_2} perform similarly in practice. Finally, D_{L_2} enjoys stronger support in standard libraries such as NumPy, enabling efficient broadcasting and further acceleration, whereas D_{W_2} lacks such optimized implementations. Together, these considerations make D_{L_2} the natural choice for the greedy search step.

The final choice concerns dW , which determines the resolution of decomposition. This effect is illustrated in Fig. 2. Smaller dW leads to a path closer to the geodesics and thereby less mass movement and fewer actions. Smaller dW also leads to simpler subtasks, which require fewer actions to accomplish, in turn allowing the greedy approach to better

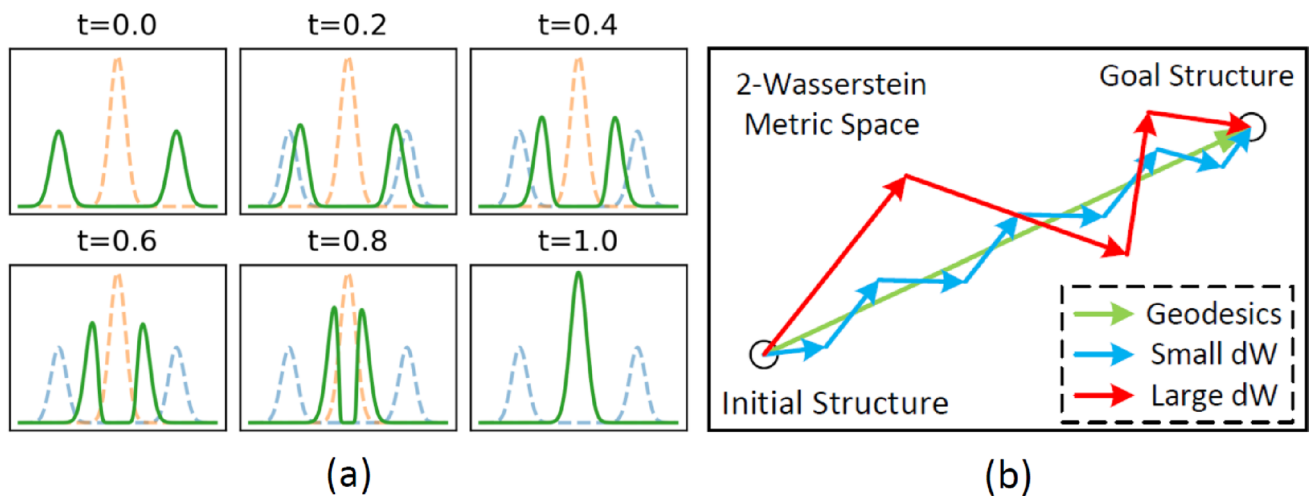


Fig. 2 **a** Wasserstein geodesics (green solid line) from h_0 (blue dotted line) to h_1 (orange dotted line). **b** Effect of dW on how Alg. 1 follows the geodesics

Table 1 Time complexity of Euclidean and 2-Wasserstein distance in 1D and 2D. The FFT Sinkhorn method is used to compute D_{W_2} in 2D to reflect the state-of-the-art result (Lakshmanan et al., 2022)

Dimension	D_{L^2} (Euclidean)	D_{W_2} (2-Wasserstein)
1D	$\mathcal{O}(L)$	$\mathcal{O}(L + M \log(L))$
2D	$\mathcal{O}(L^2)$	$\mathcal{O}(KL^2 \log L)$

L is the number of sampling points of the height function. M is the number of sampling points of the inverse cumulative distribution function in Eq. 14. K is the number of iterations in the FFT Sinkhorn method

predict the action sequence, and a more efficient algorithmic solution. The trade-off is more computation and more frequent communication with the centralized controller.

3.4 Naive greedy algorithm

As a baseline, we propose a naive greedy algorithm (Alg. 2) that directly minimizes the Euclidean distance between the current structure h and the goal structure h_1 at every step. The algorithm has the same assumption as Alg. 1 and is executable by a single robot only. At each iteration, the robot takes the action that yields the largest immediate reduction in D_{L_2} , without decomposing the task into intermediate sub-goals. While this approach is computationally efficient and simple to implement, it often becomes trapped in local minima and performs poorly in long-horizon tasks, thereby highlighting the advantages of Wasserstein-geodesics-based planning.

Algorithm 1 Single-Robot planning algorithm based on Wasserstein geodesics.

Require: construction area Q , initial height function h_0 , goal height function h_1 , desired distance to move in every subtask dW

- 1: Set current structure $h \leftarrow h_0$
- 2: **while** W decreases **do**
- 3: Get distance to the goal (Eq. 17): $W \leftarrow D_{W_2}(h, h_1)$
- 4: Set the point on the geodesics (Eq. 16): $t \leftarrow \min\{1, dW/W\}$
- 5: Get intermediate goal (Eq. 19): $h_t \leftarrow G_t[h, h_1]$
- 6: $A \leftarrow 0$
- 7: $S \leftarrow \emptyset$
- 8: **while** L decreases **do**
- 9: Find optimal action: $\phi^*, \theta^* \leftarrow \operatorname{argmin}_{\phi, \theta} D_{L_2}(h + A + f_{\phi, \theta}, h_t)$
- 10: Get distance (Eq. 18) to the intermediate goal: $L \leftarrow D_{L_2}(h + A + f_{\phi^*, \theta^*}, h_t)$
- 11: $A \leftarrow A + f_{\phi^*, \theta^*}$
- 12: Add optimal action (ϕ^*, θ^*) to S
- 13: **end while**
- 14: Execute the action sequence S (Eq. 6) and update h .
- 15: **end while**

Algorithm 2 Naive greedy algorithm.

Require: construction area Q , initial height function h_0 , goal height function h_1 .

- 1: Set current structure $h \leftarrow h_0$
- 2: **while** L decreases **do**
- 3: Get distance to the goal (Eq: 18): $L \leftarrow D_{L_2}(h, h_1)$
- 4: Find optimal action: $\phi^*, \theta^* \leftarrow \operatorname{argmin}_{\phi, \theta} D_{L_2}(h + f_{\phi, \theta}, h_1)$
- 5: Execute the action based on the discrete dynamics (Eq. 6) and update h .
- 6: **end while**

4 Centralized multi-robot algorithm

Algorithm 3 Centralized multi-robot planning algorithm based on Wasserstein geodesics. A robot swarm with N robots is identified by $\{1, \dots, N\}$. The x-coordinates of their positions are denoted by $X = [x_1, \dots, x_N]$. The reachable area and traversable area of robot i are denoted by $Q_{R,i}$ and $Q_{T,i}$, respectively.

Require: construction area Q , initial height function h_0 , goal height function h_1 , desired distance to move in every subtask dW , number of robots N , steepness threshold for taking an action to flatten the structure K_{TH}

- 1: Set current structure $h \leftarrow h_0$, and deploy robots uniformly over untraversable area Q_U
- 2: **while** $\forall i, Q_{T,i} \neq Q$ **do**
- 3: **for** each robot i **do**
- 4: Choose a random location $\tilde{\phi} \in Q_{T,i}$ that satisfies: $|h'(\tilde{\phi})| > K_{TH}$
- 5: **if** $h'(\tilde{\phi}) > K_{TH}$ **then**
- 6: $\tilde{\theta} \leftarrow \text{left}$
- 7: **else**
- 8: $\tilde{\theta} \leftarrow \text{right}$
- 9: **end if**
- 10: Execute action $\tilde{\phi}, \tilde{\theta}$ (Eq. 6) and update h and X
- 11: **end for**
- 12: **end while**
- 13: Robots spread out to uniformly cover the entire structure
- 14: **while** W decreases **do**
- 15: Get distance to the goal (Eq: 17): $W \leftarrow D_{W_2}(h, h_1)$
- 16: Set the point on the geodesics: $t \leftarrow \min\{1, dW/W\}$
- 17: Get intermediate goal (Eq: 19): $h_t \leftarrow G_t[h, h_1]$
- 18: $A \leftarrow 0, S \leftarrow \emptyset$
- 19: **while** L decreases **do**
- 20: Find optimal action: $\phi^*, \theta^* \leftarrow \operatorname{argmin}_{\phi, \theta} D_{L_2}(h + A + f_{\phi, \theta}, h_t)$
- 21: Get distance (Eq: 18) to the intermediate goal: $L \leftarrow D_{L_2}(h + A + f_{\phi^*, \theta^*}, h_t)$
- 22: $A \leftarrow A + f_{\phi^*, \theta^*}$
- 23: Add optimal action (ϕ^*, θ^*) to S
- 24: **end while**
- 25: **for** each action ϕ_j, θ_j in S **do**
- 26: **if** ϕ_j is reachable by any robot **then**
- 27: Choose execution robot i s.t. $\phi_j \in Q_{R,i}$ and $|x_i - \phi_j|$ is minimized
- 28: Execute the action (Eq. 6) and update h and X
- 29: **end if**
- 30: **end for**
- 31: **end while**

We propose a centralized multi-robot planning algorithm based on Wasserstein geodesics (Alg. 3). The algorithm follows the same assumptions of Alg. 1. Our motivation is to mitigate individual robot motion constraints and to reduce overall travel distance by assigning tasks across multiple robots. While we do not explicitly model either effect, we know from real experiments that travel distance is directly related to energy expenditure and structure degradation due to robot motion. The latter is related to the robot's speed, the structure's shape, and the granules' material properties. E.g., the change would be subtle if the robot travels on a flat surface, but pronounced if the robot climbs a steep slope.

Although following the Wasserstein geodesics minimizes the mass moving distance, it does not minimize the traveling distance of the robots that execute the construction. The traveling distance depends on how we plan for the robots to follow the Wasserstein geodesics. A good example is the construction task shown in Fig. 2a, where we want to split one mound in the middle into two mounds on each side. If we use a single robot to follow the geodesics, it will travel back and forth between two mounds increasing the travel distance. This problem can be addressed by introducing multiple (simple and therefore inexpensive) robots to work together. E.g., deploy two robots such that each robot only works on one mound.

Before introducing the algorithm, we further define traversable and untraversable areas based on the robot's motion constraints. Given robot position x_R , let $Q_T \subset Q$ denote the traversable area of a robot over which it can freely move back and forth. Q_T is the largest closed interval that includes x_R and satisfies: $\forall x \in Q_T, |h'(x)| \leq K_R$. We also define the untraversable area $Q_U \subset Q$ as any area that satisfies: $\forall x \in Q_U, |h'(x)| > K_R$.

4.1 Algorithm

In practice, the initial or goal structures often contain untraversable areas. To address this, the algorithm starts with the flattening phase (lines 2–12). In this phase, the central planner distributes robots uniformly across the untraversable

areas and lets each agent perform local leveling actions whenever the local gradient $|h'(x)|$ exceeds a predefined threshold K_{TH} . By repeatedly flattening steep regions, robots can incrementally expand their traversable areas, ensuring that the subsequent construction phase can be carried out. This phase terminates once all robots have access to the full terrain, i.e., $\forall i, Q_{T,i} = Q$.

After the flattening phase comes the construction phase (lines 14–31), where robots collaboratively follow the high-level plan decomposed via Wasserstein geodesics, similar to the single-robot case. At each subtask, the planner determines an intermediate target structure h_t along the geodesic. Next, the planner computes a candidate action sequence $S = \{(\phi_j, \theta_j)\}$ that leads to this intermediate structure. Each action is assigned to the closest robot that can reach location ϕ_j . If no robot can execute the action, it is skipped to avoid infeasible assignments. This assignment strategy ensures that actions are distributed to minimize cumulative travel distances while still progressing towards the global goal.

4.2 Limitation

A key limitation of the centralized algorithm is that, despite involving multiple robots, only one robot executes an action at a time (line 28), which limits parallelism and reduces potential efficiency gains. It is possible to break down the subtask (changing from h to h_t) into several action sequences that can be executed simultaneously to increase the parallelism. E.g., in Fig. 2a, we can break down the change from $t = 0$ to $t = 0.2$ into two action sequences (one on the left and one on the right) and let two robots execute them simultaneously. How to parallelize the change and optimally distribute actions among robots is beyond the scope of this paper and is left for future work. In addition, the communication requirements scale linearly with the number of robots, since the centralized planner must coordinate and assign every action. While this overhead is manageable for small to medium teams, it becomes a significant bottleneck for very large swarms, where communication latency and bandwidth constraints can severely impact performance.

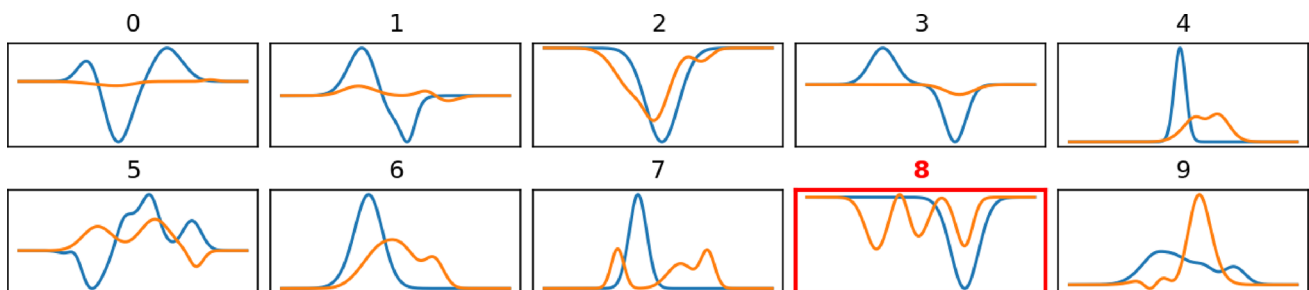


Fig. 3 Example of 10 tasks from the 100 randomly generated construction tasks. Each task comprises an initial structure h_0 (blue line) and a goal structure h_1 (orange line). Notice that for all tasks, the maximum

steepness of the initial or goal structure is close to the structure constraint K and greater than the maximum steepness the robot can climb

5 Distributed multi-robot algorithm

Algorithm 4 Distributed multi-robot coordination algorithm based on Wasserstein geodesics. A robot swarm with N robots is identified by $\{1, \dots, N\}$. The x-coordinates of their positions are denoted by $X = [x_1, \dots, x_N]$.

Require: construction area Q , initial height function h_0 , goal height function h_1 , desired distance to move in every subtask dW , number of robots N , time limit T

- 1: Set current structure $h \leftarrow h_0$, and deploy robots uniformly across the structure
- 2: Set action plan of each robot $S_i \leftarrow \emptyset$
- 3: **while** time $\leq T$ **do**
- 4: **for** each robot i **do**
- 5: **if** $S_i = \emptyset$ **then**
- 6: **if** no robot nearby **then**
- 7: Sense local structure $\tilde{h}$ (Eq. 20) and get local goal $\tilde{h}_1$
- 8: Get distance to the local goal (Eq. 17) $\tilde{W} \leftarrow D_{W_2}(\tilde{h}, \tilde{h}_1)$
- 9: Set point on the local geodesics $t \leftarrow \min\{1, dW/\tilde{W}\}$
- 10: Get local intermediate goal (Eq. 19) $\tilde{h}_t \leftarrow G_t[\tilde{h}, \tilde{h}_1]$
- 11: $A_i \leftarrow 0$
- 12: **while** L decreases **do**
- 13: Find optimal action $\phi^*, \theta^* \leftarrow \operatorname{argmin}_{\phi, \theta} D_{L_2}(\tilde{h} + A_i + f_{\phi, \theta}, \tilde{h}_t)$
- 14: Get distance to the goal (Eq. 18) $L \leftarrow D_{L_2}(\tilde{h}_t + A_i + f_{\phi^*, \theta^*}, \tilde{h}_t)$
- 15: $A_i \leftarrow A_i + f_{\phi^*, \theta^*}$
- 16: Add optimal action (ϕ^*, θ^*) to S_i
- 17: **end while**
- 18: **else**
- 19: Randomly walk on the structure
- 20: **end if**
- 21: **else**
- 22: Execute (Eq. 6) and remove an action from S_i
- 23: **end if**
- 24: **end for**
- 25: **end while**

The limitations of the centralized approach motivate the need for a distributed algorithm. In large swarms, centralized coordination quickly becomes a communication bottleneck, and sequential task execution prevents robots from exploiting their parallel capabilities. Furthermore, in remote areas, we cannot always assume access to a centralized planner with global communication and up-to-date knowledge of the terrain. A distributed strategy instead allows each robot to make decisions based on local sensing and limited interactions, enabling simultaneous actions, reduction of communication overhead, and improved scalability.

5.1 Assumptions

We assume that each robot is constrained to local measurements, i.e. can measure its current position along the horizontal axis x_r and can measure the steepness of the local structure within the range of its own length l_r . The availability of these sensor inputs matches the most recent

hardware advances for the robot in the lab, and it is reasonable to assume 1) that the actual robots have access to certain location references that ground where the construction takes place, and 2) that they can use onboard IMU data to infer the steepness of the local structure.

Given the steepness of the local structure h' , the shape of the local structure $\tilde{h}(x)$, $x \in [x_r - l_r/2, x_r + l_r/2]$ can be reconstructed:

$$\tilde{h}(x) = \int_{x_r - l_r/2}^x h'(u) du. = h(x) - h(x_r - l_r/2) \quad (20)$$

Since the robot does not have altitude information, $h(x_r - l_r/2)$ is unknown to the robot. Therefore, the inferred structure shape $\tilde{h}$ has an unknown constant offset to h . Since the robot knows its current position, it knows the corresponding local goal structure $\tilde{h}_1$.

We no longer assume long-range communication among robots, but we do assume that the robots can sense the

presence of other robots within the range of their own length l_r .

5.2 Algorithm

Our distributed multi-robot algorithm can be seen in Alg. 4. In the distributed setting, each robot operates autonomously without direct commands from a central planner. Robots are initially deployed uniformly across the structure to ensure even coverage of the workspace. After deployment, each robot repeatedly cycles through a sense–plan–act loop as follows:

1. *Initialize*: The robot begins with an empty action plan.
2. *Collision avoidance*: The robot senses nearby robots in its vicinity. If another robot is detected, it performs a random walk on the structure to avoid interference before re-evaluating. If no neighbor is present, it proceeds to infer the local shape of the terrain.
3. *Local planning*: Using its local inference, the robot computes an intermediate target structure in its neighborhood. It then generates an action plan to reach this subgoal via a greedy search, following the same principle as the single-robot algorithm (Sect. 1).
4. *Execution*: The robot executes its action until done.
5. *Repeat*: Upon completing its action plan, the robot returns to Step 1.

Notice that since there is no central controller that terminates the construction process when desired progress is reached, this loop only ends when the robot's internal timer reaches a preset time limit. This time limit is empirically set to allow enough time for convergence. Through this repeated loop, robots iteratively improve the structure in a decentralized manner. Since all decision-making is local and concurrent, the system naturally scales to large swarms while avoiding

the sequential bottlenecks of centralized control. Another advantage of using local Wasserstein geodesics is that the time complexity of computing D_{L_2} and D_{W_2} becomes independent of the overall structure size. Both distances are evaluated only within a fixed local window l_r . Consequently, the computational cost per robot remains constant, regardless of the global structure length, which greatly relaxes the computational requirements for each robot.

5.3 Limitation

The distributed algorithm also has several limitations. First, congestion can arise in narrow workspaces where robots cannot maintain enough isolation distance. This congestion can lead to idle time while they randomly move until no robot is nearby, reducing the efficiency. Second, although robots use local Wasserstein geodesics to guide their actions, these do not capture the global Wasserstein geodesic of the structure. As a result, local decisions may diverge from the globally optimal modification path, requiring longer convergence or leading to failed construction. These limitations are discussed in details in Sect. 6.3. Finally, in reality, the inference of local terrain shape is subject to sensor noise (not modeled here), which can reduce accuracy in planning and execution. Together, these limitations highlight the trade-off between scalability and precision in multi-robot planning.

6 Evaluation

To evaluate the algorithmic performance, we generated random structures based on a mixed Gaussian model to form a set of construction tasks. A construction task is defined by a tuple (Q, h_0, h_1) where Q is the construction area and h_0, h_1 are valid height functions of initial and goal structures defined on Q . The random height function is given by:

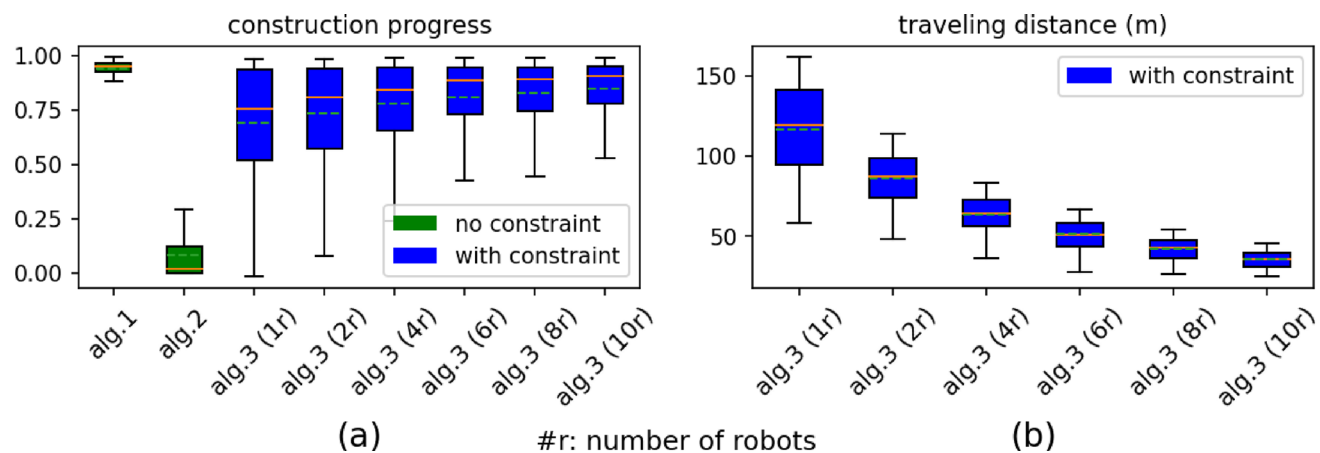


Fig. 4 Simulation results of solving the set of 100 construction tasks. Each task is simulated 5 times. **a** Construction progress. **b** Normalized robot traveling distance throughout the construction

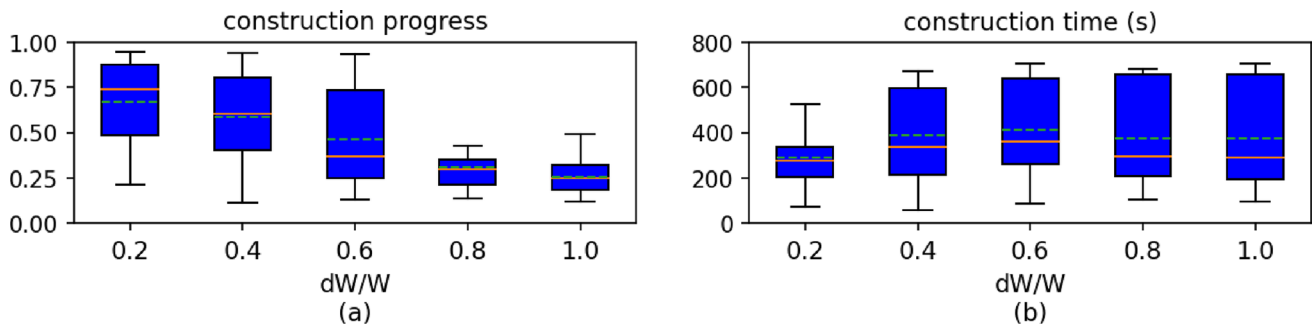


Fig. 5 Simulation results of solving task #8 (highlighted in red in Fig. 3). Each setting of dW is simulated 100 times. W is the distance between the initial and goal structure. **a** Construction progress vs dW . **b** Normalized construction time vs dW

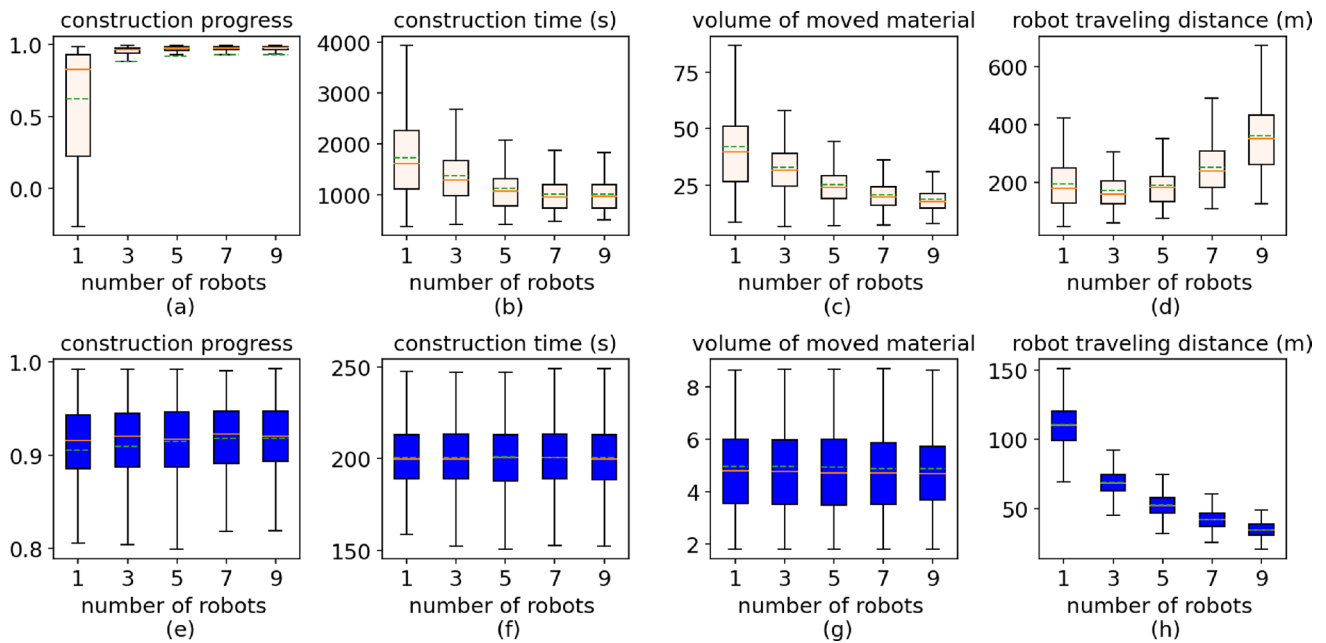


Fig. 6 Simulation results of constructing random tasks with traversable initial and goal structures by Alg. 4 (shown in **a–d**) and Alg. 3 (shown in **e–h**). For each boxplot, the median is an orange solid line and the mean is a green dashed line. **a, e** Achieved construction progress. **b,**

f Normalized construction time. **c, g** Normalized robot traveling distance throughout construction. **d, h** Normalized volume of moved material throughout construction

$$h(x) = \sum_{i=1}^m a_i e^{-\frac{(x-b_i)^2}{2c_i^2}}, \quad (21)$$

where $m \in \mathbb{Z}$, $a_i \in \mathbb{R}$, $b_i \in \mathbb{R}^+$, $c_i \in \mathbb{R}$ are random variables drawn from uniform distributions. We generated 100 construction tasks on a fixed Q that is 5 m (13 times the robot length) wide. 20 randomly selected tasks are shown in Fig. 3. These tasks do not have practical purposes, but are used to test the cross-task performance of the algorithm.

To test the construction performance of the algorithm, we use several metrics:

- *Achieved construction progress* is measured as shown in Sect. 3.2.

- *Success rate* is the number of successful simulations divided by the total number of simulations (500 in our case) per task. We define that a construction task is successfully completed when 90% construction progress is reached.
- *Construction time* is the time to reach 90% structure completion. Therefore, it can only be measured for the ones that reach at least 90% of the goal. To eliminate the effect of the variance in the difficulty and size of random tasks, we normalize the construction time using the Wasserstein distance between the initial and goal structure.
- *Robot traveling distance* is the total distance robots traveled to complete construction. It is normalized in the same way as the construction time.

- *Volume of moved material* is the total volume of material moved by robots throughout a successful construction process, which is an important indicator of energy expenditure and construction efficiency. To eliminate the structural variance, we normalize it using the volume difference between the initial and goal structure $\int_Q |h_0(x) - h_1(x)| dx$, which can be seen as the ideal least volume of material needed to be moved to complete the construction.

6.1 Single-robot algorithm

We used the 100 construction tasks to evaluate Algs. 1–2. The average Wasserstein distance between the initial and goal structure is around 7. Therefore, we set dW in Alg. 1 to 1 for an appropriate ratio of subtask to overall task. Each task is simulated 5 times.

We first compare the simulation results shown in Fig. 4 for Alg. 1 and Alg. 2. We temporarily remove the robot constraints from Sect. 2.4 to show the algorithms' performance in an ideal setting. The average progress achieved by Alg. 1 and Alg. 2 is $94 \pm 3\%$ and $8 \pm 12\%$, respectively. This shows that greedily following locally optimal actions often cannot solve the full construction problem. It also shows that without robot constraints, the Wasserstein-geodesics-based algorithm can accomplish almost any construction task within the physical limits of the granular material. The average number of steps necessary to complete 95% of a structure is 2000 ± 1185 with each step taking 1 s. Since the construction area is only 5 m wide, the large number of actions shows the long-horizon nature of the construction problem, which again strengthens the importance of subtask abstraction. It also suggests that other methods with poor sampling efficiency, like reinforcement learning, may be infeasible.

Adding in the robot constraints (Fig. 4a) show their strong adverse effects: the average progress drops by 27%.

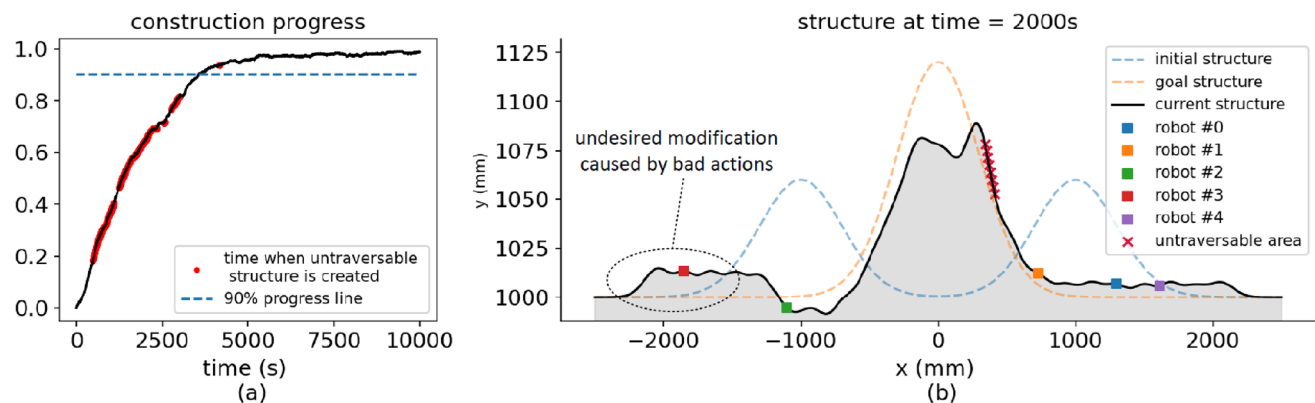


Fig. 7 **a** Construction progress over time for the task in Fig. 2a by Alg. 4 on 5 robots. The moment when the untraversable structure is created is marked by a red cross. **b** The state of construction at a moment when bad actions cause an untraversable structure (highlighted in the dashed circle)

6.2 Centralized multi-robot algorithm

The same setup in Sect. 6.1 is applied to test Alg. 3.

Figure 4a shows that as more robots are added, Alg. 3 achieves higher progress. The highest average progress is $85 \pm 13\%$ for ten robots, a 23% increase from the single-robot case. To study the effect of the number of robots on the traveling distance, we execute Alg. 3 with motion constraint and analyze 25 tasks with construction progress $> 90\%$. Figure 4b shows that as more robots are added, the total traveling distance of all robots decreases. The average traveling distance drops from $116 \pm 25\text{m}$ for one robot to $35 \pm 5\text{m}$ for ten robots, a 70% decrease. We also found that the initial traveling distance for flattening the structure is only up to 8% of the total traveling distance, which is almost negligible.

Under the influence of the motion constraint, the effect of dW is amplified as the algorithm has a much higher chance of solving simpler subtasks. We study this effect by solving task #8 (highlighted in red in Fig. 3), which has the lowest progress, even when being built by multiple robots. The distance between the initial and goal structure of task #8 is $W = 15.39$. We solve this task with 10 robots and dW ranging from $0.2W$ to W . When $dW = W$, the algorithm always attempts to solve the task in one shot. Figure 5 shows that as dW increases, the progress drops, and more actions are taken. This illustrates the importance of subtasking when solving the long-horizon construction problem.

6.3 Distributed multi-robot algorithm

We found that the distributed algorithm does not work well with untraversable initial or goal structures. Therefore, in this section, we focus only on traversable initial and goal structures. We tested Alg. 3 and Alg. 4 with the 100 construction tasks that have traversable initial and goal structures, and a construction area of 5 m (13 times the robot

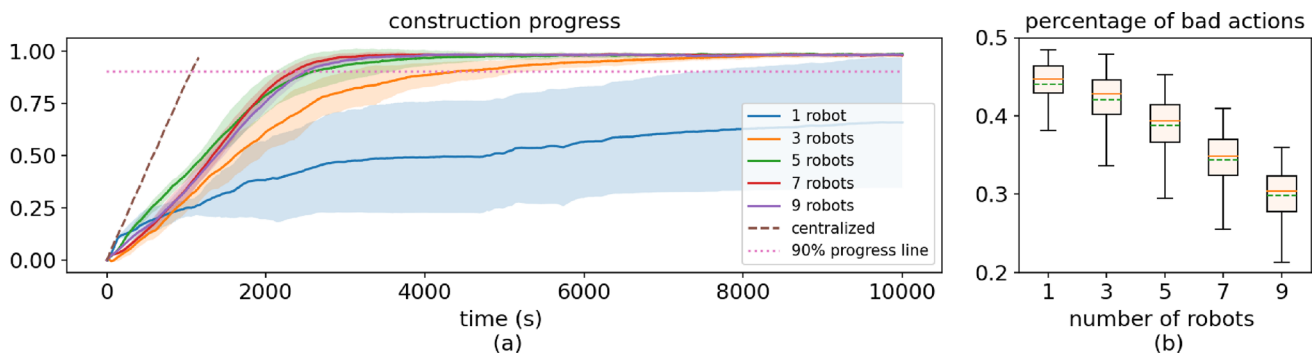


Fig. 8 **a** Construction progress over time for the task in Fig. 2a by Alg. 4 with different numbers of robots (solid line) and by Alg. 3 (dashed line). Each setting is simulated 10 times. **b** Percentage of bad

actions among all actions for construction random tasks with traversable initial and goal structures by Alg. 4. Medium is the orange solid line, and mean is the green dashed line

length) wide. The average Wasserstein distance between the initial and goal structures is around 5 globally and around 1 locally (within the range of the robot's length). This is expected since the global difference is always greater than the local difference. Therefore, we set $dW = 1$ for the centralized algorithm and $dW = 0.25$ for the distributed algorithm for an appropriate ratio of subtask to overall task. Each task is simulated 5 times. The time limit T is set to 10,000 time steps (10,000 s).

Figure 6a–d shows simulation results for the distributed algorithm. The success rate is 40%, 83%, 88%, 91%, and 90% for 1, 3, 5, 7, and 9 deployed robots, respectively. Figure 6a shows that when three or more robots are deployed, the construction progress does not change significantly. However, to achieve a similar construction progress, Fig. 6b, c shows it takes less time and requires less material to be moved when more robots are deployed. Compared with the single-robot case, deploying 7 robots increases the average construction progress from $62 \pm 37\%$ to $93 \pm 17\%$ (50% increase), and reduces the construction time and volume of moved material from $1731 \pm 786\text{m}$ to $1019 \pm 367\text{m}$ (70% decrease) and 42 ± 21 to 21 ± 6 (50% decrease), respectively.

One drawback of deploying more robots is the crowding effect shown in Fig. 6d. As more robots are deployed, the traveling distance increases. Since robots tend to spread out to avoid interference, when the construction site gets too crowded, robots travel more to avoid each other.

Figure 6e–h shows comparative simulation results for the centralized algorithm. The success rate is 63%, 65%, 67%, 69%, and 70% for 1, 3, 5, 7, and 9 robots, respectively. Although the success rate is low, the overall progress is still over 90%. Figure 6e shows that when more than one robot is deployed, the distributed algorithm reaches a similar construction progress at $92 \pm 4\%$, but has a higher success rate. The lower success rate compared with the distributed algorithm is likely caused by the early stopping when the algorithm could not find a way to move towards the goal

(Alg. 3 (line 17)). The reduction in traveling distance as more robots are deployed is seen in Fig. 6h, consistent with the observation in Sect. 6.2. In general, the centralized algorithm is much more efficient and stable in terms of construction time, traveling distance, and volume of moved material due to the more optimal planning based on global information and perfect communication.

A major reason why the distributed algorithm fails is the untraversable structures created during construction. Even though both the initial and goal structures are traversable, an untraversable structure can still emerge due to the limited local information available to the robots. An example is shown in Fig. 7. Figure 7a shows that an untraversable structure can emerge at any time throughout the construction process. Upon the occurrence, part of the structure may be temporarily inaccessible for some robots, which may cause the construction to stall. The direct cause of this phenomenon is the 'bad action' that emerges when the local intermediate goal structure directs the robot to push material in a globally suboptimal direction. There is no bad action for the centralized algorithm since it always makes decisions based on global Wasserstein geodesics. An example is shown Fig. 7b, where robots attempt to combine two mounds on each side to form a larger mound in the center (the task in Fig. 2a). In the area circled by the black dashed line, a robot created a flat structure to match the shape of the goal locally. By doing so, the robot took a bad action, pushing material left instead of towards the center, causing a decrease in progress.

The remedy for this issue is to deploy more robots. When more robots are deployed, local feedback between the robots, mediated through the shared structure, better approximates the global Wasserstein geodesics used by the central algorithm. This can be seen in Fig. 8, which shows that as more robots are deployed, the time evolution of construction progress of the distributed algorithm approaches that of the centralized algorithm, and the percentage of bad actions also decreases by 32%, from $44 \pm 3\%$ to $30 \pm 3\%$

on average. This is also reflected in the occurrences of untraversable structures. Among all simulations, 82%, 72%, 67%, 68%, and 62% simulations create untraversable structures for 1, 3, 5, 7, and 9 robots, indicating that deploying more robots reduces the likelihood of creating untraversable structures. Out of those simulations where untraversable structures are created, 35%, 77%, 84%, 87%, and 85% of the simulations are successful, indicating that deploying more robots helps overcome untraversable structures, similar to the effect of the centralized algorithm. The reduced likelihood of taking bad actions and creating untraversable structures explains why the construction process is more stable and rises to completion more quickly when there are more robots.

7 Conclusion

This paper introduced a novel framework for collective terrain modification by minimalistic earthmover robots using Wasserstein-based planning. We presented an abstract model of the construction process, planning algorithms based on Wasserstein geodesics, and both centralized and distributed multi-robot implementations. Through extensive simulations, we demonstrated that our planner yields more reliable and scalable performance than the greedy baseline. Key results include: 1) for the centralized algorithm, deploying more robots can increase the average progress by 23% and reduce the traveling distance by 70%, and 2) for the distributed algorithm, deploying more robots can increase the average progress by 50%, and reduce the construction time and volume of moved material by 70% and 50%, respectively. Both suggest that the construction process converges more quickly and stably as swarm size grows.

To our best knowledge, this is the first work to apply optimal transport theory and Wasserstein geodesics to collective robotic construction. Beyond the immediate application of shaping terrain with robot swarms, our framework has broader implications for robotics and automation. The formulation of collective modification as an optimal transport problem offers a transferable strategy for other continuous-material tasks, such as manipulating deformable media. More generally, the use of geometry-aware planning in swarm systems highlights an approach where large numbers of simple robots can achieve results that traditionally requires more complex robotics, thereby widening the scope of scalable, resilient, and low-cost robotic solutions.

7.1 Future work

Despite its advantages, our framework has several limitations. For the centralized algorithm, only one robot acts at

a time, and communication scales linearly with swarm size, creating a bottleneck for very large teams. For the distributed algorithm, congestion in narrow workspaces can lead to idle time, and local Wasserstein geodesics may diverge from global objectives. These limitations are consistent with the failure modes observed in our simulations (Sect. 6.3). Specifically, distributed robots occasionally produced untraversable structures and took locally reasonable but globally suboptimal actions, both of which stalled progress. While larger swarms helped overcome these effects, the results highlight that without global coordination, distributed systems remain vulnerable to congestion and misaligned local actions. More broadly, our simulations assume simplified physics, which leaves a gap to real-world deployment.

Future research will focus on overcoming these limitations. Promising directions include hybrid strategies that combine local distributed rules with occasional global updates, energy-aware planning that jointly minimizes robot travel and material displacement, and congestion-aware task allocation to improve scalability. A key step is extending the framework to 3D terrain modification. The current formulation of the problem and the proposed algorithms naturally generalize to 3D. The next step is to implement high-efficiency numerical methods for solving the optimal transport problem, such as FFT Sinkhorn (Lakshmanan et al., 2022). Finally, hardware validation with minimalist robot prototypes on granular media will be critical to bridging the gap between simulation and field deployment.

Author contributions Jiahe Chen developed the method implementation and carried out the experiments. All authors wrote and reviewed the manuscript.

Funding This project was funded by the Packard Fellowship for Science and Engineering and the National Science Foundation (NSF) Grant #2042411.

Data availability Data is provided within the manuscript or supplementary information files.

Code availability Code will be released at: <https://github.com/Ericland/collective-terrain-modification-2d>.

Declarations

Conflict of interest There is no conflict of interest relevant to the content of this article.

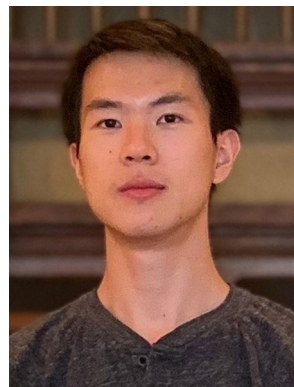
References

- Angenent, S., Haker, S., & Tannenbaum, A. (2003). Minimizing flows for the Monge-Kantorovich problem. *SIAM Journal on Mathematical Analysis*, 35(1), 61–97.

- Arjovsky, M., Chintala, S., & Bottou, L. (2017). Wasserstein generative adversarial networks. In *International conference on machine learning* (pp. 214–223). PMLR.
- Chartrand, R., Wohlberg, B., Vixie, K., & Bollt, E. (2009). A gradient descent solution to the Monge-Kantorovich problem. *Applied Mathematical Sciences*, 3(22), 1071–1080.
- Cheraghi, A. R., Shahzad, S., & Graffi, K. (2022). Past, present, and future of swarm robotics. In *Intelligent systems and applications: Proceedings of the 2021 intelligent systems conference (IntelliSys)* (Vol. 3, pp. 190–233). Springer.
- Cuturi, M. (2013). Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in Neural Information Processing Systems*, 26, 65.
- Desai, P., Galica, C. A., & Werkheiser, N. (2022). Nasa lunar surface innovation initiative: Ensuring a cohesive, executable strategy for technology. In *International astronautical congress*.
- Hong, S., Bangunharcana, A., Park, J.-M., Choi, M., & Shin, H.-S. (2021). Visual slam-based robotic mapping method for planetary construction. *Sensors*, 21(22), 7715.
- Hong, S., Shyam, P., Bangunharcana, A., & Shin, H. (2022). Robotic mapping approach under illumination-variant environments at planetary construction sites. *Remote Sensing*, 14(4), 1027.
- Huang, L. R., Zhu, A., Wang, K., Goldman, D. I., Ruina, A., & Petersen, K. H. (2022). Construction and excavation by collaborative double-tailed saw robots. *IEEE Robotics and Automation Letters*, 7(2), 3742–3748.
- Kolouri, S., Park, S. R., Thorpe, M., Slepcev, D., & Rohde, G. K. (2017). Optimal mass transport: Signal processing and machine-learning applications. *IEEE Signal Processing Magazine*, 34(4), 43–59.
- Lakshmanan, R., Pichler, A., & Potts, D. (2022). Nonequispaced fast Fourier transform boost for the Sinkhorn algorithm. arXiv preprint [arXiv:2201.07524](https://arxiv.org/abs/2201.07524)
- Lee, R., Younes, B., Pletta, A., Harrington, J., Wong, R. Q., & Whitaker, W. (2023). Cratergrader: Autonomous robotic terrain manipulation for lunar site preparation and earthmoving. arXiv preprint [arXiv:2311.01697](https://arxiv.org/abs/2311.01697)
- Lellmann, J., Lorenz, D. A., Schonlieb, C., & Valkonen, T. (2014). Imaging with Kantorovich-Rubinstein discrepancy. *SIAM Journal on Imaging Sciences*, 7(4), 2833–2859.
- Monge, G. (1781). Mémoire sur la théorie des déblais et des remblais. *Memories of mathematical and physics academic royale science* (pp. 666–704).
- Moses, R. W., & Mueller, R. P. (2021). Requirements development framework for lunar in situ surface construction of infrastructure. In *Earth and Space 2021* (pp. 1141–1155).
- Napp, N., & Nagpal, R. (2014). Distributed amorphous ramp construction in unstructured environments. *Robotica*, 32(2), 279–290.
- Oberman, A. M., & Ruan, Y. (2015). An efficient linear programming method for optimal transportation. arXiv preprint [arXiv:1509.03668](https://arxiv.org/abs/1509.03668)
- Ondras, J., Ni, D., Deng, X., Gu, Z., Zheng, H., & Bhattacharjee, T. (2022). Robotic dough shaping. In *2022 22nd International conference on control, automation and systems (ICCAS)* (pp. 300–307). IEEE.
- Petersen, K. H., Napp, N., Stuart-Smith, R., Rus, D., & Kovac, M. (2019). A review of collective robotic construction. *Science Robotics*, 4(28), 8479.
- Ramdas, A., García Trillos, N., & Cuturi, M. (2017). On Wasserstein two-sample testing and related families of nonparametric tests. *Entropy*, 19(2), 47.
- Saboia, M., Thangavelu, V., & Napp, N. (2019). Autonomous multi-material construction with a heterogeneous robot team. *Robotics and Autonomous Systems*, 121, Article 103239.
- Shachaf, D., Inbar, O., & Zarrouk, D. (2019). Rsaw, a highly reconfigurable wave robot: Analysis, design, and experiments. *IEEE Robotics and Automation Letters*, 4(4), 4475–4482.
- Thangavelautham, J., & Xu, Y. (2022). The design of autonomous robotic technologies for lunar launch and landing pad (llp) preparation. In: *2022 IEEE aerospace conference (AERO)* (pp. 1–13). IEEE.
- Thangavelu, V., Silva, M. S., Choi, J., & Napp, N. (2020). Autonomous modification of unstructured environments with found material. In *2020 IEEE international conference on robotics and automation (ICRA)* (pp. 7798–7804). IEEE.
- Zarrouk, D., Mann, M., Degani, N., Yehuda, T., Jarbi, N., & Hess, A. (2016). Single actuator wave-like robot (saw): Design, modeling, and experiments. *Bioinspiration & Biomimetics*, 11(4), Article 046004.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Jiahe Chen received the Ph.D. degree in Electrical and Computer Engineering from Cornell University in 2024, where he worked with Prof. Kirstin Petersen in the Collective Embodied Intelligence Lab. He received his M.Sc. in Electrical Engineering from the University of Pennsylvania in 2019 and his B.Sc. in Engineering Physics from Queen's University at Kingston in 2017. His research focuses on decentralized algorithms for multi-robot construction systems, emphasizing scalability and fault tolerance. He will be joining General Motors (GM) as a Senior Applied Scientist, working on robotics foundational models.



Kirstin Petersen is an Associate Professor and Aref and Manon Lahham Faculty Fellow in the School of Electrical and Computer Engineering at Cornell University. Her work is focused on design and coordination of robot collectives able to achieve complex behaviors beyond the reach of an individual, and corresponding studies on how social insects do so in nature. Major research topics include swarm intelligence, embodied intelligence, soft robots, and bio-hybrid systems. Petersen did her postdoc at the Max Planck Institute for Intelligent Systems and her PhD at Harvard University and the Wyss Institute for Biologically Inspired Engineering.